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The Yin and Yang of Arnt2 in Activity-Dependent Transcription

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Spatiotemporal regulation of neuronal gene expression is essential for proper functioning of neuronal circuits. In this issue of *Neuron*, Sharma et al. (2019) discover a dual role for Arnt2-NcoR2 protein complexes in the activity-dependent regulation of neuronal transcriptomes.

The rapid modification of cellular transcriptomes represents a fundamental mechanism for neuronal homeostasis and plasticity. During development, activity-dependent gene expression programs coordinate the functional wiring of neuronal microcircuits. In the mature brain, activity-dependent programs enable homeostatic scaling, for example, by elevating neuronal inhibition in cells exposed to excess excitation (Turrigiano, 2008). Lastly, in learning and memory, activity-dependent gene transcription might support the strengthening of connections between neurons that are active during encoding, thereby creating ensembles with spatiotemporal activity patterns that can be recreated during memory retrieval (Josselyn et al., 2015). Over the past decades, there has been major progress on the identification of gene regulatory elements and transcription factors driving the onset of activity-dependent transcripts in neurons. However, comparably little is known about mechanisms that repress target gene transcription prior to stimulation. Tight repression would minimize noise from

background expression and thus maximize the dynamic range that can be accomplished in response to stimulation. But how can robust repression be achieved without compromising the dynamics of gene induction in response to stimulation? Conceivably, a rapid onset of activation is critical to associate transcriptional responses with a particular set of stimuli. In this issue of *Neuron*, Sharma and colleagues (Sharma et al., 2019) now uncover transcription factor complexes that facilitate rapid switching from repression to activation of transcription in rodent neurons.

The spatiotemporal control of transcription across the entire genome is coordinated by a complex interplay of transcription factors, their binding sites in promoters and enhancers, and the accessibility of these binding sites. Long-term silencing of regulatory elements is frequently directed by DNA methylation, which alters chromatin structure and thus reduces access for transcriptional regulators (Lister et al., 2013). However, in the mature brain, there are alternative mechanisms for silencing that are more rapidly

reversible upon neuronal stimulation. At activity-dependent promoters, proteins that modify histone composition or histone modifications have important functions in silencing transcription prior to stimulation. For example, the nucleosome remodeler complex NuRD triggers inactivation of promoters (Yang et al., 2016) or recruitment of the histone deacetylase HDAC4 represses genes in inactive neurons (Sando et al., 2012). Nonetheless, the spatiotemporal regulation of the full repertoire of activity-dependent gene regulatory elements in neuronal cells—in particular at the level of enhancers—is poorly understood.

Sharma and colleagues (Sharma et al., 2019) now uncover novel mechanisms underlying the switch from repression prior to neuronal activity to gene activation with sensory stimulation in mouse neurons (Figure 1). They focus on the transcription factor Npas4, a neuron-specific, activity-induced immediate early gene, which binds promoters and enhancers and regulates transcription of late response genes in a cell-type-specific manner (Spiegel et al., 2014). For example, an Npas4-dependent

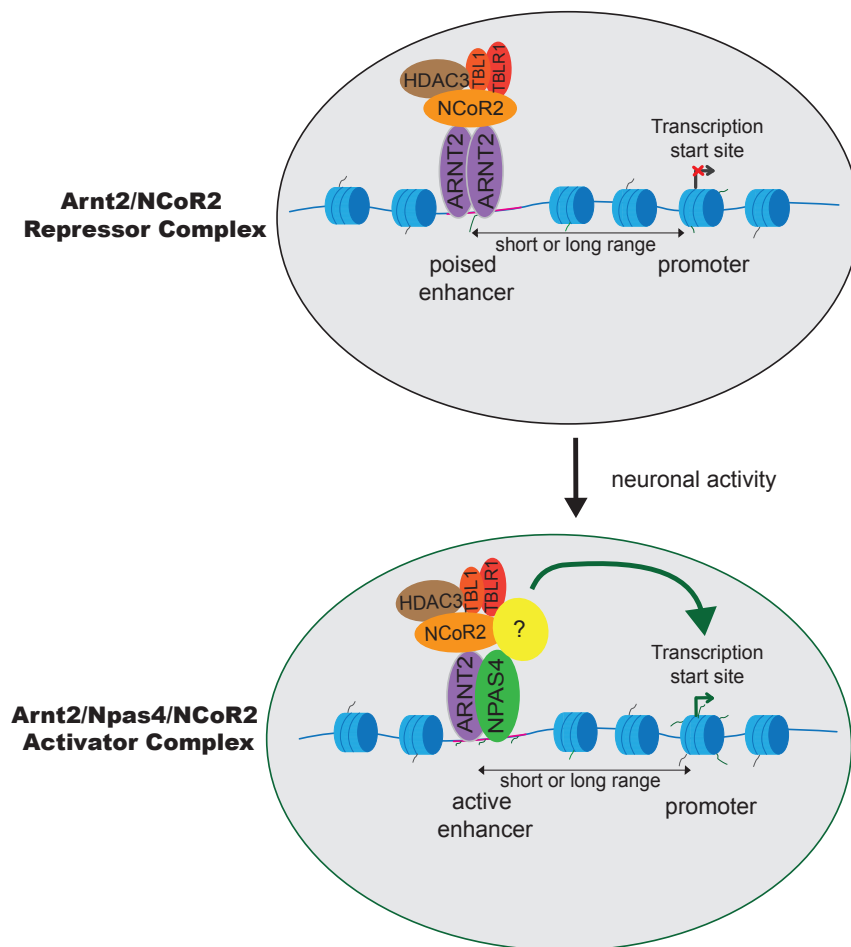


Figure 1. Neuronal Activity-Dependent Switch from Repression to Activation

Model for the repression of neuronal activity-dependent genes at enhancers by Arnt2-NCoR2 complexes (NCoR2, HDAC3, TBL1, and TBLR1). Upon reaching a threshold of neuronal stimulation, Arnt2 homodimers are converted to Arnt2-Npas4 heterodimers. The resulting complex containing NCoR2 and presumably additional activators triggers transcription of Npas4-dependent, activity-regulated genes.

transcriptional program drives the elevation of perisomatic inhibition in hippocampal pyramidal cells upon neuronal activation, and loss of Npas4 in CA3 pyramidal cells impairs contextual memory formation in mice (Weng et al., 2018). These critical functions of the Npas4-dependent program raise the question of how inappropriate de-repression of Npas4 transcripts prior to sensory stimuli is achieved.

Sharma et al. (2019) first performed genome-wide mapping of Npas4 binding sites by ChIP-seq (chromatin immunoprecipitation sequencing). Then they combined these data with ATAC-seq (assay for transposase-accessible chromatin sequencing) and H3K27ac (acetylation of histone 3 lysine 27) ChIP-seq mapping

to identify accessible chromatin regions and regulatory elements recruited in stimulated neurons. They demonstrate that a majority of Npas4-bound regions are distal regulatory enhancer elements and that activation of select enhancers is Npas4 dependent. Interestingly, the chromatin of Npas4-regulated enhancer elements is accessible even prior to sensory stimulation. This suggested that these regions are already bound by additional transcription factor complexes rather than requiring more extensive remodeling of chromatin structure for activation.

Npas4 is a basic helix-loop-helix Per-Arnt-Sim (bHLH-PAS) domain protein family member (Ooe et al., 2004). Sharma et al. (2019) hypothesized that additional bHLH-PAS proteins may be bound to reg-

ulatory elements prior to Npas4 recruitment. Based on a survey of single-cell sequencing data, Sharma et al. (2019) identified the transcription factor Arnt2 as the most significantly expressed neuronal bHLH-PAS domain family member as a candidate for occupying Npas4 binding sites in silent neurons. Genome-wide analyses confirmed that Arnt2 binding regions largely overlap with the Npas4-related regulatory elements. Importantly, loss of Arnt2 resulted in significant de-repression of Npas4-dependent, activity-regulated transcripts. Moreover, Arnt2 loss of function phenocopied functional neuronal alterations observed upon elevation of Npas4 protein levels. These experiments support the model that prior to stimulation, activity-dependent genes are actively repressed by a repressor complex and that upon neuronal stimulation, this repression is converted into transcriptional activation. Yet, Arnt2 itself is not a transcriptional repressor. Thus, Sharma et al. (2019) went on to identify interactors of Arnt2 that might mediate transcriptional repression. They discovered as an Arnt2 interaction partner the so-called NCoR2 co-repressor complex that has well-established roles as repressor of transcription in several cell types (Mottis et al., 2013). NCoR2 binding regions largely cover the Arnt2-bound regulatory elements, strongly suggesting that Arnt2 indeed recruits NCoR2 for the repression of activity-dependent transcripts prior to stimulation.

So, if Arnt2-NCoR2 complexes mediate repression, how is this repression converted into transcriptional activation upon neuronal stimulation? Interestingly, in stimulated cells, the Arnt2- and Npas4-bound regions overlap. This means that Arnt2 itself is not removed from these sites upon stimulation. Instead, Arnt2 forms heterodimers with Npas4, which then drive transcription of Npas4-target genes upon neuronal stimulation. How the Npas4/Arnt2 heterodimers activate transcription remains to be discovered. Surprisingly, Npas4/Arnt2 still recruit the NCoR2 complex, indicating that NCoR2 may switch from its canonical repressor function to drive transcriptional activation (Figure 1).

The study from Sharma et al. (2019) reveals a dual role for Arnt2 in the repression and activation of Npas4-regulated

neuronal transcripts. This study establishes distal enhancer elements as major contributors to activity-dependent neuronal gene regulation. Interestingly, the chromatin of these Npas4 target sites is largely accessible for transcription factor binding, presumably facilitating rapid gene induction upon stimulation. The extensive functional analysis for the transcription factor Arnt2 demonstrates a powerful repression mechanism that prevents aberrant transcription of activity-dependent genes and hence dysregulation of neuronal circuits. Some fascinating questions remain: additional work will be needed to identify how the Arnt2-NCoR2 repressor complex is turned into an activator complex and engages with Npas4 for the transcription of Npas4-dependent genes. This functional conversion may involve posttranslational modifications and/or an interplay between NCoR2 and chromatin remodeling complexes.

Another exciting extension of this work will be the exploration of the cell-type-specific interpretation of activity-dependent stimuli. Npas4 binds thousands of regulatory elements across the entire genome. Previous work demonstrated that Npas4-dependent transcriptional programs significantly differ between neuronal cell types (Spiegel et al., 2014). This selectivity is thought to emerge during development where a cell-type-specific subset of regulatory elements is poised for future transcriptional control

by Npas4. Such mechanisms may generally enable broadly expressed transcription factors to produce cell-type-specific readouts. Thus, the complex interplay of binding sites, their developmental specification, and the cell-type-specific complement of transcription factors are likely to account for the divergent gene regulation processes in different neuronal cells. Notably, Sharma et al. (2019) find Arnt2 to interact with Npas3, another bHLH-PAS transcription factor. Npas3 expression is enriched in inhibitory interneurons, thus providing one additional piece of the puzzle of how cell-type-specific responses might be controlled. Ultimately, this work greatly advances our insight into how tightly controlled, rapid induction of neuronal transcript dynamics can be induced by sensory stimulation. The findings pave the way for an exploration of cell-type-specific transcriptional mechanisms, their roles in neuronal plasticity and cognitive function, and how they may contribute to neuronal dysfunction in disease states.

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